Lesson 2

Household Wiring Systems

Objective: You will be able to:

- Describe wiring system in house.
- Connect switch, fuse, and circuit breaker.
- Connect a light bulb with two switches
- Understand safety rule.

1-Electrical Power Transmission

This electric current is generated in a power plant, and then sent out over a power grid to your homes, and ultimately to your power outlets.



When you plug in an electric kettle, you are connecting it into a circuit. The circuit hasn't got a battery in it. But the mains supply is doing much the same job.

The current from a mains socket isn't a one-way flow like the current from a battery. Instead, it is pushed and pulled forwards and backwards round the circuit 50 times every second known as frequency.

The current is known as **alternating current** or **AC**. The **mains frequency** is 50 Hz.

Power stations supply **AC** because it is easier to generate than one-way **direct current (DC)**. In Cambodia, the supply voltage is 220V in your house.

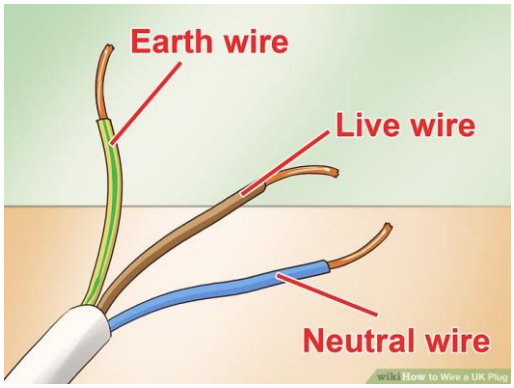
2-Electrical Devices

Live wire is the wire that is connected directly to high voltage (main line) and delivers current to appliances.

It has the voltage about 220 Volts to your house and has the color of **red** or **brown**.

Neutral wire is the wire that complete the circuit by providing a return path to the supply for current.

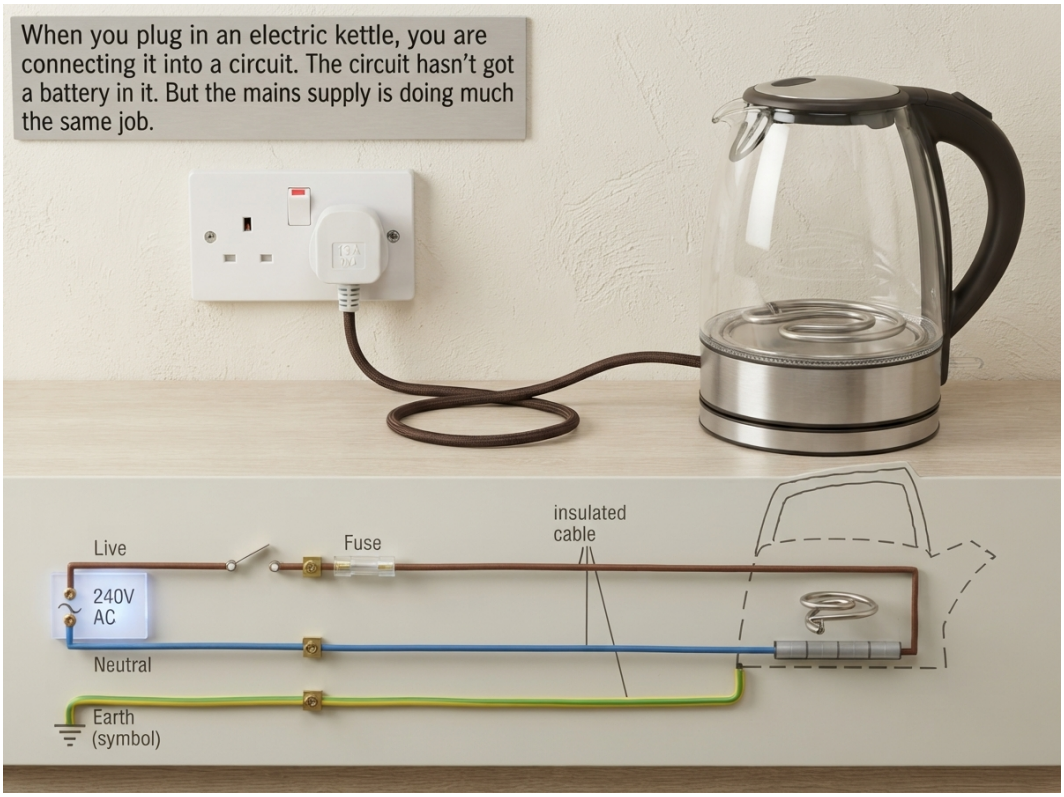
It is earthed by the electricity board at the generator. The Neutral wire is at approximately zero volts. Its color is **blue**. But you must NEVER touch this wire either!!



Earth wire is a low resistance wire that is connected from the metal body of the appliances to Earth. Its color is **green**-**yellow**.

For normal function, it carries no electric current. The earth wire gives a safe route for the current if the live wire touches the outer casing.

It stops the appliances ever becoming 'live'. *Results* – a large current flows to Earth.



How does an earth wire work?

When the metal casing becomes 'live' accidentally, the earth wire will allow current to flow from the casing to the earth. Since the path has low resistance, the current will be large and that will blow the fuse and isolate the appliance.

The switch is a device use to turn on or off the electrical devices.

The switch must be connected to the live wire so that the appliance will not be 'live' (or connected to a high potential) when the switch is open.

The fuse is a short piece of thin wire which overheats and melts when too much current flows through it. If a fault develops, the fuse 'blows' and breaks the circuit before anything else can overheat and catch fire.

Fuse must be connected to live wire.

Example: Plugs are normally fitted with 3 A or 13 A fuses. The value tells you the current needed to 'blow' the fuse.

If a TV takes a current of 0.5 A. its plug should be fitted with a 3A fuse.



3A, 5A and 13A fuses used in 3-pin plugs

If a kettle takes a current of 10A, its plug should be fitted with a 13A fuse.

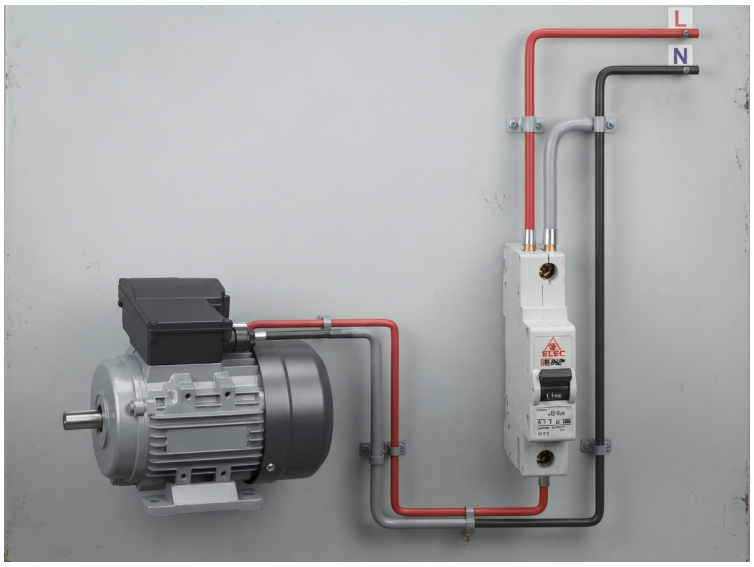
The fuse value should always be more than the actual current, but as close to it as possible. The TV will still work with a 13A fuse fitted. But it might not be safe. If something goes wrong, the circuits could overheat and catch fire, without the fuse blowing.

When the current exceeds the rated value, a wire inside the fuse will melt to break the circuit hence protecting the appliance.

A **circuit breaker** is an electromagnetic device that breaks a circuit when the current goes above a certain value.

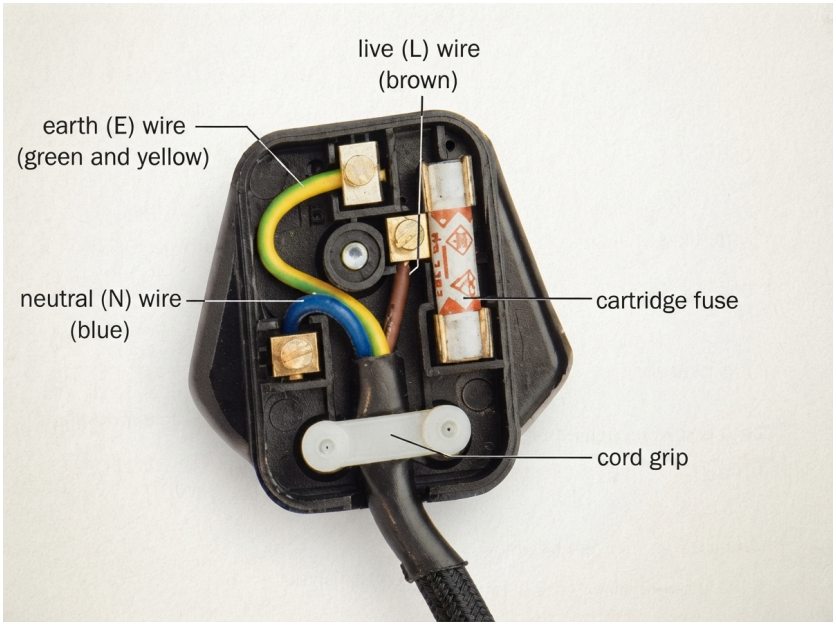






Three pins plug: A fused plug connects an electrical appliance to the mains supply via the power socket.

The cartridge fuse inside the plug protects the appliance when there is an electrical fault. If excessive current flows in the appliance, the fuse blows. This breaks the circuit and isolates the appliance with the fault, so that overheating does not damage it.



Double insulation

Some electrical appliances use two pins plug instead of three-pin plug. For such appliances, there is no earth wire. These appliances use double insulation to protect user from electric shocks.

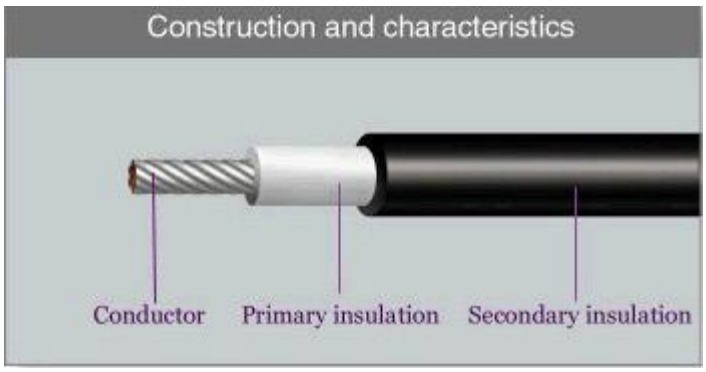
Double insulation is a safety feature that can replace the earth wire. Appliances that have double insulation usually use a two pins plug, as only the live and neutral wires are applied.

Double insulation provides two level of insulation:

1-The electric cables are insulated from the internal components of appliance.

2-The internal components are insulated from the external casing Appliances with double insulation typically have non-metallic casing such as plastic.

Hand-held electrical appliance, such as hair dryer, electric power drill, are protected by double insulation.







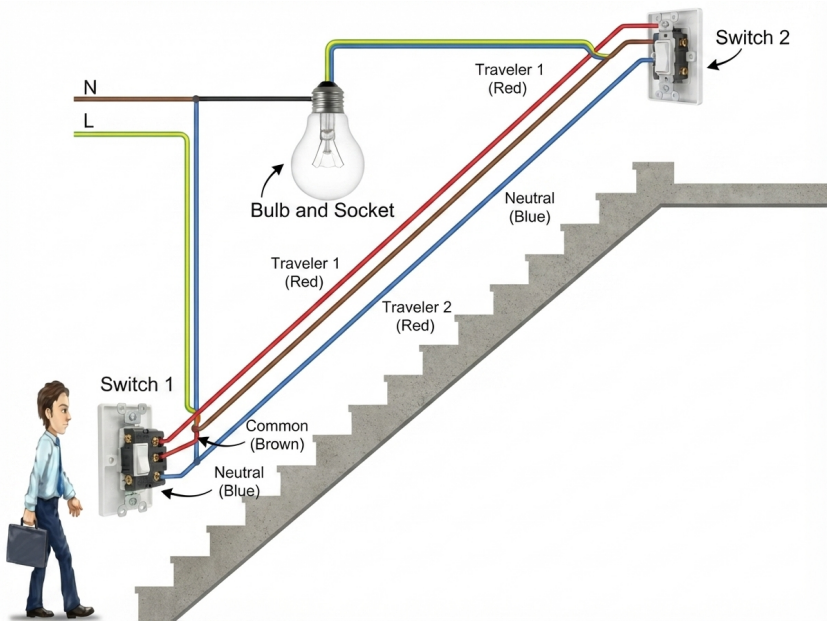
THE PRINCIPLE OF DOUBLE INSULATION

Such appliances do not have metal casing or conducting part in contact with the user. Hence, **no earth wire** connection is provided.

3-Two ways switches

In most houses, you can turn the landing light on or off using upstairs or downstairs switches. These aren't the simple 'on-off' type. They have two contacts instead of one. They are **two-way switches**.

If the switches are both up or both down, then the current flows through the bulb. But if one switch is up and the other down, the circuit is opened. Move either switch and you reverse the effect of the other one.



5-Dangers of electricity

Electric shock

Electric shock occurs if current flows from an electric circuit through a person's body to earth.

This can happen if there is damaged insulation or faulty wiring. The typical resistance of dry skin is about 10 000 Ω, so if a person touches a wire carrying electricity at 240 V, an estimate of the current flowing through them to earth would be I = V/R =240/10 000 = 0.024 A = 24 mA. For wet skin, the resistance is lowered to about 1000 Ω (since water is a good conductor of electricity) so the current would increase to around 240 mA. It is the size of the current (not the voltage) and the length of time for which it acts which determine the strength of an electric shock. The path the current takes influences the effect of the shock; some parts of the body are more vulnerable than others. A current of 100 mA through the heart is likely to be fatal.



To avoid the risk of getting an electric shock:

• Use plugs that have an earth pin and a cord grip; a rubber or plastic case is preferred.

• Do not allow appliances or cables to come into contact with water.

• Do not have long cables trailing across a room, under a carpet that is walked over regularly or in other situations where the insulation can become damaged.

In case of an electric shock, take the following action:

1.Switch off the supply if the shocked person is still touching the equipment.

2.Send for qualified medical assistance.

3.If breathing or heartbeat has stopped, commence CPR (cardiopulmonary resuscitation) by applying chest compressions at the rate of about 100 a minute until there are signs of chest movement or medical assistance arrives.

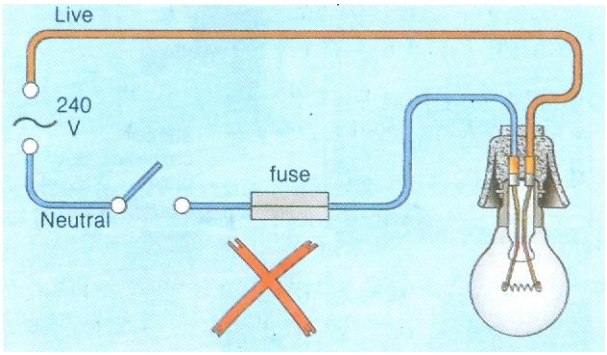
Questions

1. Why must the switch be connected to the live wire?

2. Copy and complete the table to show whether a 3A ,5A or 13A fuse should be fitted to the plug connected to each appliance. The first is done for you.

Appliance	Current takes in A	Fuse value in A
Radio	0.1	3
Hairdryer	4	
Refrigerator	0.5	
Cassette player	0.2	
Fan heater	12	
Food mixer	2	

This circuit has been wrongly wired.



If the bulb is taken out of its socket, the circuit isn't safe. Explain why not. Redraw the circuit, showing the correct wiring.

4. How does the fuse work?

5. Why must the switch be connected to the live wire?

6. How does an earth wire work?

7. Which wire (earth, live and neutral) has the highest potential?

8. What is the function of a fuse?

10. What is the function of an earth wire?

11. Most electrical wire is covered in a rubber or plastic coating called insulation. What is the purpose of having this "insulation" covering the metal wire?

12-What is the difference between a fuse and a circuit breaker?

13.Explain why birds do not get shocked when they perch on a power line, even if both of their feet touch the wire. Explain why birds become electrocuted if their wings bridge between two different power conductors:

